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MESH-BFT: Diversity-Weighted Post-Quantum Byzantine Fault Tolerance for Directed Acyclic Meshes

Author: Nenad Vasic

Abstract. We present MESH-BFT, a Byzantine fault tolerant consensus design for Directed Acyclic Meshes (DAMs) that combines three elements. (1) *Diversity-weighted settlement*: correlated witnesses are discounted via a multi-dimensional correlation function, so a Byzantine coalition whose identities share an organization, subnet or location keeps less effective stake than its raw stake. The discount does not raise the classical one-third bound against an adversary that presents distinct profiles, and in the current implementation it is reported as a confirmation level rather than gating finality. (2) *Post-quantum signatures* under a formally defined quantum polynomial-time adversary model: Profile A records carry two signatures (ML-DSA-65 and SPHINCS+-SHA2-192f), so record authorship stays unforgeable if one scheme is broken; witness attestations, and therefore settlement, currently carry a single ML-DSA-65 signature. (3) *Per-record causal finality* in a constant number of attestation rounds, independent of the total number of records, so causally concurrent records finalize in parallel. We prove safety assuming Byzantine stake below one third, prove post-quantum unforgeability of Profile A records, give a proof sketch of seal liveness that the current implementation does not yet meet, and model-check bounded instances of the settlement rule in TLA+. We report measurements from a six-node geo-distributed testnet (March–April 2026, since retired) with gossip latency $p50 = 250$ ms. The public network currently runs with a single staked identity, the genesis authority, so it tolerates no Byzantine stake today (Section 8.1). To our knowledge, MESH-BFT is among the first designs to combine diversity-weighted attestation, post-quantum signatures and per-record finality in one framework.

Status of this edition (2026-09-25). A self-audit of this paper against the reference implementation found claims that were false or unsupported. This edition corrects them: (1) Theorem 1 now states its assumption that Byzantine stake is below one third of the eligible stake; diversity weighting does not raise that bound. (2) Theorem 2 covers record authorship only: witness attestations carry a single ML-DSA-65 signature, and the composite security level is about the stronger scheme’s (192 bits), not the sum (384). (3) Durable per-record finality uses raw 2/3 attesting stake; diversity weighting is computed and reported but does not gate it. (4) Seal settlement (Theorem 4) is not live in the current implementation, and the seal-rank beacon can be ground by the preceding sealer, so the $(1/3)^7$ bound does not hold against such a sealer. (5) The hop-count formula of Theorem 3 was algebraically wrong and is restated with an explicit fan-out. (6) The evaluation now matches the six-node dataset, and Section 8.1 describes the current single-staker network.

Each change is marked “Correction (2026-09-25)” where it applies; the reference implementation’s known gaps are listed in docs/KNOWN-LIMITATIONS.md of the public repository (github.com/navigatorbuilds/elara-mesh).

1. Introduction

Byzantine fault tolerant (BFT) consensus is fundamental to distributed systems, from state machine replication [1] to blockchain protocols [2, 3]. Classical BFT protocols (PBFT [1], HotStuff [2]) assume independent node failures and impose total ordering on all operations, creating leader bottlenecks and $O(n)$ or $O(n^2)$ communication per decision.

DAG-based protocols (DAG-Rider [4], Narwhal/Tusk [5], Aleph [6]) improve throughput by building a directed acyclic graph of proposals and deriving total order from the graph structure. However, they still assume independent Byzantine failures and require global total ordering.

To our knowledge, three gaps remain open in the literature:

Gap 1: Correlated failures. Real-world validators are not independent. Cloud outages, software bugs, and organizational control create correlated failure patterns. Ethereum’s client diversity research [14] and the GPoS protocol [9] recognize this problem empirically, but no formal BFT safety proof incorporates validator diversity as a parameter.

Gap 2: Post-quantum security. NIST standardized post-quantum signature schemes (ML-DSA, SLH-DSA) in 2024 [15], but no consensus protocol formally models a quantum adversary and proves BFT safety under it. DAG-Rider [4] achieves post-quantum safety as a side effect of its hash-based design, without formal quantum adversary analysis.

Gap 3: Per-record finality. Block-based consensus serializes all transactions into blocks, even when they are causally independent. Sui Lutris [7] provides per-object finality for single-writer objects, but shared-state transactions still require full consensus. No protocol proves $O(1)$ finality per record in a general DAG structure.

MESH-BFT addresses these three gaps in one framework, with the limits stated in each section. Our contributions:

1. **Settlement safety and diversity weighting (Theorem 1).** We define a multi-dimensional correlation function over witness attributes (organization, network topology, geography) and a diversity-weighted settlement predicate, and prove that no two conflicting records can both settle while Byzantine stake is below one third of the eligible stake. We show that the diversity discount reduces the effective stake of a coalition whose identities share a profile; it does not raise the one-third bound against an adversary that presents distinct profiles.
2. **Post-quantum record authorship (Theorem 2).** We define a quantum polynomial-time (QPT) adversary model and prove that Profile A records, signed with two schemes based on different hardness assumptions, stay unforgeable if either scheme is broken, at a level of about $\max(\lambda_d, \lambda_s)$. Witness attestations currently carry one ML-DSA-65 signature, so consensus under a break of ML-DSA-65 is not covered.
3. **Per-record causal finality (Theorem 3).** We show that records in a DAM reach confirmation in $O(1)$ attestation rounds, each taking $O(\log n / \log f)$ gossip hops with per-node fan-out f , with causally concurrent records finalizing in parallel. In an analytical model, throughput scales linearly with the number of independent witness clusters; this has not been measured at scale.

The remainder of this paper is organized as follows. Section 2 defines the system model. Section 3 proves settlement safety and analyzes diversity weighting. Section 4 proves post-

quantum unforgeability of Profile A records. Section 5 analyzes per-record finality and gives a proof sketch of seal liveness. Section 6 presents empirical evaluation. Section 7 discusses related work. Section 8 discusses limitations, including the current single-staker network, and future work.

2. System Model

2.1 Directed Acyclic Mesh

Definition 1 (DAM). A Directed Acyclic Mesh is a tuple $M = (V, E, Z, \tau, \sigma)$ where:

- V is a set of records (vertices)
- E subset of $V \times V$ is a set of directed edges (parent $\rightarrow$ child), forming a DAG
- $Z: V \rightarrow \{0, 1, \dots, N_z - 1\}$ is a zone assignment function over a dynamic zone count N_z (Definition 2)
- $\tau: V \rightarrow \mathbb{R}^+$ is a timestamp function
- $\sigma: V \rightarrow \Sigma^*$ is a signature function mapping records to their cryptographic signatures

Definition 2 (Zone Assignment). For record r with identifier $\text{id}(r)$:

$$Z(r) = \text{u64}(\text{SHA3-256}(\text{id}(r))[0..8]) \bmod N_z$$

where $[0..8]$ denotes the first 8 bytes interpreted as a big-endian unsigned 64-bit integer, and $N_z \geq 1$ is the network's current zone count. N_z is dynamic: it scales with network load under the zone auto-scaler (anchor-attested zone-transition records; hysteresis-damped). Early drafts of this paper fixed $N_z = 256$ and used only the first hash byte; the shipped runtime (`src/network/zone.rs::for_record_dynamic`) uses the full-entropy 8-byte form above, and retains a 256-zone form solely as a legacy-compatibility path for pre-dynamic-zone records — itself now computed as the full-entropy 8-byte value mod 256 (`for_record`), not from the first hash byte, so the historical first-byte assignment survives only in archived drafts.

Properties:

- *Deterministic:* same record always maps to same zone (for a given N_z ; zone-count transitions re-route only future records, at an anchor-attested target epoch)
- *Uniform:* SHA3-256 output is uniformly distributed, and $2^{64} \bmod N_z$ bias is negligible for any practical N_z
- *Not creator-proof:* the creator chooses $\text{id}(r)$ (UUID v7: a timestamp plus random bits the creator draws; the node also accepts a client-supplied id), so it can re-draw the identifier until the record lands in a chosen zone, at an expected cost of about N_z hash evaluations. For honest creators the assignment is uniform. (Correction 2026-09-25: earlier editions called zone assignment non-manipulable.)

Definition 3 (Causal Dependency). Record r_1 causally precedes r_2 (written $r_1 \rightarrow r_2$) iff:

- (r_1, r_2) in E , or
- there exists r_3 in V such that $r_1 \rightarrow r_3$ and $r_3 \rightarrow r_2$ (transitive closure)

Records r_1 and r_2 are **causally concurrent** (written $r_1 \parallel r_2$) iff neither $r_1 \rightarrow r_2$ nor $r_2 \rightarrow r_1$.

Property (Parallel Finality). If $r_1 \parallel r_2$, their finality processes are independent. No ordering constraint exists between them.

2.2 Identities and Cryptographic Profiles

Definition 4 (Identity). An identity is a tuple $I = (pk_d, pk_s, sk_d, sk_s, PoW, \epsilon, \pi)$ where:

- pk_d, sk_d in $\{0,1\}^*$ are ML-DSA-65 (FIPS 204) public/secret keys
- pk_s, sk_s in $\{0,1\}^*$ are SPHINCS+-SHA2-192f public/secret keys (Profile A only; the pre-standard NIST round-3 scheme, not FIPS 205 SLH-DSA)
- $PoW = (\text{nonce}, \text{difficulty})$ such that $\text{leading_zeros}(\text{SHA3-256}(pk_d || \text{nonce})) \geq \text{difficulty}$
- ϵ in $\{\text{Human}, \text{AI}, \text{Device}, \text{Organization}, \text{Composite}\}$ is entity type
- π in $\{A, B, C\}$ is cryptographic profile

Naming note (2026-09-25). The second scheme is SPHINCS+-SHA2-192f, the pre-standard NIST round-3 submission. FIPS 205 standardized a modified version as SLH-DSA, and signatures are not interchangeable. Earlier editions called the scheme SLH-DSA.

Definition 5 (Identity Hash). The identity hash is:

$$H(I) = \text{hex}(\text{SHA3-256}(pk_d))$$

Definition 6 (Cryptographic Profiles).

- **Profile A (dual-signature):** Signs with both ML-DSA-65 AND SPHINCS+-SHA2-192f. Verification requires both signatures valid.
- **Profile B (single-signature):** Signs with ML-DSA-65 only. Used by browser/constrained nodes.
- **Profile C (delegated):** Uses a parent identity's key via delegation record.

Definition 7 (Signature Verification).

$$\text{Verify_A}(r, I) = \text{Verify_DIL}(\text{sigma_d}(r), \text{signable}(r), pk_d) \text{ AND } \text{Verify_SPH}(\text{sigma_s}(r), \text{signable}(r), pk_s)$$
$$\text{Verify_B}(r, I) = \text{Verify_DIL}(\text{sigma_d}(r), \text{signable}(r), pk_d)$$

where $\text{signable}(r)$ is the canonical deterministic byte representation of r .

2.3 Witness Sets and Attestation

Definition 8 (Witness Set). For record r , the witness set $W(r)$ is the set of identities that have signed attestation records for r :

$$W(r) = \{ I : \text{there exists attestation } a \text{ such that } a.\text{record_id} = \text{id}(r) \text{ AND } \text{Verify}(a, I) = \text{true} \}$$

Definition 9 (Stake Weight). For identity I with staked balance $s(I)$:

$$w(I) = s(I) / S_{\text{total}}$$

where $S_{\text{total}} = \sum_{I \in I_{\text{all}}} s(I)$ is the total staked supply across all identities.

Definition 10 (Attesting Stake). For record r :

$$AS(r) = \sum_{I \in W(r)} s(I)$$

2.4 Diversity Function

Definition 11 (Witness Profile). A witness profile is a tuple $P(I) = (\text{org}, \text{subnet}, \text{geo})$ where:

- org in Strings — organizational affiliation
- subnet in Strings — IP subnet prefix
- geo in Strings — geographic zone

Definition 12 (Pairwise Correlation). For witnesses I_i, I_j with profiles P_i, P_j :

$\text{corr}(I_i, I_j) = \alpha * 1[\text{org}_i = \text{org}_j] + \beta * 1[\text{subnet}_i = \text{subnet}_j] + \gamma * 1[\text{geo}_i = \text{geo}_j]$
 where $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.1$ are correlation weights, and $1[.]$ is the indicator function.

Implementation note (constants, 2026-07-02). The worked examples and tables in this paper use the original specification weights ($\gamma = 0.1$, so the maximum pairwise correlation is 0.9). The shipped runtime (`src/network/consensus.rs`) uses **$\gamma = 0.2$** (maximum pairwise correlation therefore up to 1.0) together with a geographic-bucket ramp that scales γ up smoothly as a zone gains geographic diversity, rather than a single fixed value. This changes the *numbers* in the tables below but not the *direction of the result*: a larger correlation weight produces a larger diversity discount, so a correlated attacker’s effective stake shrinks at least as fast as the tables show. Every safety bound derived here holds under the shipped constants a fortiori. The shipped-constant columns are included alongside the spec-constant numbers in the Section 3.4 table, and the Theorem 1 remark after Step 4 restates the safety bound under $\text{corr} = 1.0$; read the spec-constant numbers as the conservative (weaker-discount) case.

Properties:

- $\text{corr}(I_i, I_j)$ in $[0, 0.9]$ under the paper’s spec constants (up to 1.0 under the shipped $\gamma = 0.2$; maximum when all three attributes match)
- $\text{corr}(I_i, I_j) = \alpha + \beta = 0.8$ when either profile is unknown (conservative: an unregistered witness is treated as maximally correlated short of shared geography — assumed same org and same subnet. This penalizes non-disclosure and prevents a Sybil from gaining independence credit by withholding its profile; treating unknowns as *independent* would let an unprofiled Sybil claim full independence and defeat the diversity mechanism)
- Symmetric: $\text{corr}(I_i, I_j) = \text{corr}(I_j, I_i)$

Definition 13 (Independence Factor). For witness I_n relative to witness set W :

$$d(I_n, W) = 1 / (1 + \text{Sum}_{\{I_m \text{ in } W, m \neq n\}} \text{corr}(I_n, I_m))$$

Properties:

- $d(I_n, W)$ in $(0, 1]$
- $d = 1$ when I_n has zero correlation with all others (fully independent)
- $d \rightarrow 0$ as I_n becomes increasingly correlated with more witnesses

Definition 14 (Trust Score). For record r with witness set $W(r)$:

$$T(r) = 1 - \text{Product}_{\{I_n \text{ in } W(r)\}} (1 - w(I_n) * d(I_n, W(r)))$$

Properties:

- $T(r)$ in $[0, 1]$
- $T = 0$ when $W(r) = \text{empty set}$
- T approaches 1 as more independent, well-staked witnesses attest
- Correlated witnesses contribute diminishing marginal trust

Definition 15 (Effective Stake). The diversity-adjusted attesting stake for record r :

$$\text{ES}(r) = \text{Sum}_{\{I_n \text{ in } W(r)\}} s(I_n) * d(I_n, W(r))$$

This is the key quantity for the diversity analysis of Section 3.2. $\text{ES}(r) \leq \text{AS}(r)$, with equality only when all witnesses are fully independent.

2.5 Confirmation Levels and Settlement

Definition 16 (Confirmation Levels). For record r :

```

level(r) = Anchored    if level(r) >= Confirmed AND r in epoch_seal
                        AND challenge_window(r) elapsed with no open challenge
      = Confirmed      if |W(r)| >= 3 AND clusters(W(r)) >= 3 AND settled_d(r)
      = Attested       if |W(r)| >= 1
      = Unconfirmed    otherwise

```

where $\text{settled}_d(r)$ is the diversity-weighted settlement predicate of Definition 19 (over the eligible-stake denominator of Definition 18), and challenge_window is 24 hours in the shipped runtime.

Implementation note (confirmation gate). Earlier drafts modeled the third Confirmed conjunct as a tunable scalar trust threshold, $T(r) \geq \text{tau_confirm}$. No such parameter exists in the shipped runtime: the third conjunct is the settlement predicate itself (`src/network/consensus.rs`). Three shipped behaviors refine this definition: (i) *Small-network fallback* — when witness profiles are unknown or fewer than 3 distinct organizations are profiled, the diversity-weighted predicate falls back to raw 2/3 attesting stake (Definition 18) over the same denominator; otherwise diversity weighting is mandatory. (ii) *Cross-zone acceleration (shared-witness bridging)* — when ALL of r 's cross-zone parents are already finalized, the cluster requirement relaxes from 3 to 1 for r . The witness-count and stake thresholds are never relaxed; the rationale is that the causal chain has already been independently verified in another zone. (iii) *Anchored challenge window* — inclusion in an epoch seal alone does not yield Anchored status; a 24-hour challenge window must elapse with no open equivocation challenge against the record. (iv) *Level names and promotion* — the shipped runtime names the levels Pending (Unconfirmed above), Sealed (Attested), Finalized (Confirmed) and Anchored (ConfirmationLevel, `src/network/consensus.rs`), and promotes a record to Finalized when raw per-record settlement (Definition 18) passes — see the correction after Definition 19.

Definition 17 (Cluster Detection). Witnesses are partitioned into clusters via union-find:

1. Merge l_i, l_j if $\text{org}(l_i) = \text{org}(l_j)$ AND $\text{subnet}(l_i) = \text{subnet}(l_j)$
2. Merge l_i, l_j if $|\text{tau}(\text{att}_i) - \text{tau}(\text{att}_j)| \leq \text{Delta_timing}$ (500ms timing correlation)
3. Unknown-profile witnesses start as their own cluster, merged only by timing

$\text{clusters}(W)$ = number of distinct roots after union-find

Definition 18 (Settlement, amended). Record r is settled iff:

$$\text{AS}(r) * 3 \geq \text{S_elig}(r) * 2$$

where the eligible-stake denominator is

$$\text{S_elig}(r) = \text{S_D}(Z(r)) - s(\text{creator}(r))$$

and $\text{S_D}(z)$ is the settlement denominator for zone z : the registered per-epoch committee stake for z when a committee is active, otherwise the full zone stake (with long-silent stake aged out under liveness decay, floor-guarded). The creator's stake is excluded because a creator is forbidden from attesting its own record — this is enforced at ingest (self-attestations are rejected at insert and retroactively purged), so creator stake is unreachable for the record's quorum and leaving it in the denominator would make the 2/3 threshold unreachable for records created by high-stake identities. The shipped runtime cites this form as “Definition 18, amended” (`src/network/consensus.rs::is_settled`).

Definition 19 (Diversity-Weighted Settlement). Record r achieves diversity-weighted settlement iff:

$$\text{ES}(r) * 3 \geq \text{S_elig}(r) * 2$$

Using effective stake (diversity-adjusted) instead of raw attesting stake, over the same eligible-stake denominator as Definition 18. This is strictly stronger: correlated witness sets must accumulate more raw stake to meet the threshold. (The shipped runtime evaluates this predicate in deterministic fixed-point arithmetic — $Q = 10^9$ — rather than floating point, so nodes that hold the same attestation set and ledger state reach bit-identical settlement verdicts regardless of architecture; `is_settled_diverse`, `docs/SETTLEMENT-FLOAT-DETERMINISM.md`.)

Correction (2026-09-25) — what gates finality. Durable per-record finality requires attesting stake of at least $2/3$ of the zone’s eligible stake, excluding the creator (Definition 18). Diversity weighting (Definition 19) is reported as a confirmation level; it does not gate finality in the current implementation. Earlier editions described the diversity-weighted predicate as the finality gate. Two further gaps are tracked as implementation fixes: attestations that arrive before their record are buffered and later admitted without the minimum-stake and identity-age checks, so the admitted set can depend on arrival order; and when a zone committee is active, the numerator counts every attester while the denominator is committee stake. Both are latent in today’s single-staker network (Section 8.1).

2.6 Adversary Models

Definition 20 (Classical Byzantine Adversary). An adversary A_c that:

- Controls $f < n/3$ identities (by stake weight)
- Can deviate arbitrarily from the protocol
- Can delay messages but not prevent eventual delivery (partial synchrony)
- Cannot forge signatures under the computational hardness of ML-DSA-65

Definition 21 (Correlated Byzantine Adversary). An adversary A_{corr} that:

- Controls f identities, all sharing the same profile P (same org, subnet, geo)
- Raw stake weight: $\sum s(I)$ for I in controlled set
- Effective stake weight: $\sum s(I) * d(I, W)$ where d is diminished by correlation

Key insight: Under diversity-weighted settlement, A_{corr} with stake f needs MORE raw stake than A_c to achieve the same effective stake, because correlated attestations are discounted — provided its identities are recorded with correlated profiles. Profiles are self-reported or inferred (Section 8.1), so an adversary can avoid the discount by presenting distinct profiles.

Definition 22 (Quantum Polynomial-Time Adversary). An adversary A_q that:

- Has access to a quantum computer capable of running Shor’s algorithm
- Can break ONE of $\{\text{ML-DSA-65, SPHINCS+-SHA2-192f}\}$ with non-negligible probability
- Cannot break BOTH simultaneously (different hardness assumptions: lattice vs hash-based)
- All other capabilities same as A_c

Definition 23 (Sybil Adversary). An adversary A_s that:

- Can create unlimited identities, each requiring PoW cost $\geq 2^{\text{difficulty}}$
- Each identity must stake $\geq \text{MIN_WITNESS_STAKE}$ (100 beats — the internal, non-tradeable protocol credit) to participate in consensus
- Entity clustering with diminishing returns: $\text{earning_rate}(N) = \log_{10}(1 + N)$

3. Settlement Safety and Diversity Weighting

3.1 Theorem Statement

Theorem 1 (Settlement Safety). Let r and r' be conflicting records (the same signed (creator, nonce) slot, Section 3.2.1) whose settlement is evaluated against the same eligible-stake denominator S_{elig} (Definition 18), where every attester counted in the numerator belongs to the stake counted in S_{elig} . Assume the total stake of Byzantine identities within that eligible stake is less than $S_{\text{elig}}/3$, and that honest identities do not attest conflicting records (Section 3.2.1). Then r and r' cannot both settle, under either the raw predicate (Definition 18) or the diversity-weighted predicate (Definition 19).

Correction (2026-09-25). Earlier editions stated Theorem 1 without an assumption on Byzantine stake and claimed that diversity weighting raises the Byzantine threshold “beyond the classical $n/3$ bound”, so that a correlated adversary “cannot achieve double settlement regardless of stake”. That is false as stated. An adversary holding a third or more of the eligible stake can double-settle by using one identity, or identities recorded with distinct profiles, which the discount does not touch; the bundled TLA+ model spec/tla/MCTightBreak.tla exhibits a conflicting finalization when two of four equal-stake, fully independent witnesses are Byzantine. The earlier proof analyzed only a coalition that shares one profile, which is the case where the discount is largest — the defender’s best case, not the worst. Theorem 1 now states the one-third assumption and proves safety by quorum intersection; the earlier Steps 1–5 are kept below as an analysis of what the discount does to a shared-profile coalition.

3.2 Proof

By Definition 15, $ES(r) \leq AS(r)$, so a record that meets the diversity-weighted predicate (Definition 19) also meets the raw predicate (Definition 18). It therefore suffices to show that r and r' cannot both settle under the raw predicate.

Suppose both do:

$$AS(r) \geq (2/3) * S_{\text{elig}}$$

$$AS(r') \geq (2/3) * S_{\text{elig}}$$

Every attester counted in $AS(r)$ or $AS(r')$ belongs to the eligible stake, so the stake of identities that attested both records is at least

$$AS(r) + AS(r') - S_{\text{elig}} \geq (1/3) * S_{\text{elig}} > S_B$$

where S_B is the total Byzantine stake within the eligible stake. At least one identity that attested both records is therefore honest, which contradicts the assumption that honest identities do not attest conflicting records. ■

Scope. The proof needs r and r' to face the same denominator, with every counted attester inside it. Two properties of the current implementation fall outside that hypothesis; both are latent in today’s single-zone, single-staker network (Section 8.1) and tracked as implementation fixes: the settlement zone is derived from the record identifier, so two conflicting records can fall in different zones with different denominators; and with an active zone committee, the numerator counts every attester while the denominator is committee stake.

Analysis: effect of the diversity discount on a correlated coalition (not part of the safety proof). The steps below, kept from earlier editions, compute how much effective stake a Byzantine coalition keeps when all of its identities are recorded with one shared profile. They are written with S_{total} as the denominator for readability; the same algebra holds with S_{elig} .

Step 1: Partition identities.

Let H subset of I be the set of honest identities and B subset of I be the set of Byzantine identities, with $|B|$ controlled by the adversary. Define:

$$\begin{aligned} S_H &= \text{Sum}_{\{I \text{ in } H\}} s(I) && \text{(total honest stake)} \\ S_B &= \text{Sum}_{\{I \text{ in } B\}} s(I) && \text{(total Byzantine stake)} \\ S_{\text{total}} &= S_H + S_B \end{aligned}$$

Step 2: Bound effective honest stake.

For honest identity I_h with no correlation to other honest identities (best case for the honest set):

$$d(I_h, W) \leq 1$$

Therefore: $ES_H(r) \leq S_H$ for any record r .

For honest identities that ARE correlated (e.g., two honest nodes from the same organization):

$$d(I_h, W) < 1$$

Therefore: $ES_H(r) < S_H$ in general. Honest diversity matters too.

Step 3: Bound effective Byzantine stake.

The adversary controls B identities. If all Byzantine identities are recorded with the same profile (the case where the discount is largest — the defender's best case, since the adversary chooses its profiles):

$$\text{corr}(I_i, I_j) = \alpha + \beta + \gamma = 0.9 \quad \text{for all } i, j \text{ in } B$$

For Byzantine identity I_b in witness set W where all other Byzantine identities also appear:

$$\begin{aligned} d(I_b, W) &= 1 / (1 + \text{Sum}_{\{j \neq b, j \text{ in } B \text{ intersect } W\}} \text{corr}(I_b, I_j)) \\ &= 1 / (1 + 0.9 * (|B \text{ intersect } W| - 1)) \end{aligned}$$

For $|B \text{ intersect } W| = k$ Byzantine witnesses from the same profile:

$$d(I_b, W) = 1 / (1 + 0.9(k-1))$$

The total effective Byzantine stake in W :

$$ES_B(r) = \text{Sum}_{\{I_b \text{ in } B \text{ intersect } W(r)\}} s(I_b) * d(I_b, W(r))$$

If all k Byzantine identities have equal stake $s_b = S_B/k$:

$$\begin{aligned} ES_B(r) &= k * (S_B/k) * 1/(1 + 0.9(k-1)) \\ &= S_B / (1 + 0.9(k-1)) \end{aligned}$$

Key result: As k grows (more sybil identities), ES_B shrinks. The adversary's effective stake diminishes as $1/k$ for large k — for a coalition whose identities share one profile.

Step 4: Settlement overlap argument.

For record r to achieve diversity-weighted settlement:

$$ES(r) = ES_H(r) + ES_B(r) \geq (2/3) * S_{\text{total}}$$

The honest identities attesting r contribute at most S_H effective stake (less if correlated). The Byzantine identities contribute at most $ES_B \leq S_B / (1 + 0.9(k-1))$.

For BOTH r and r' to achieve settlement:

$$\begin{aligned} ES_H(r) + ES_B(r) &\geq (2/3) * S_{\text{total}} \\ ES_H(r') + ES_B(r') &\geq (2/3) * S_{\text{total}} \end{aligned}$$

Since honest identities do not attest both r and r' (by assumption), the honest witnesses in $W(r)$ and $W(r')$ are disjoint sets. Therefore:

$ES_H(r) + ES_H(r') \leq S_H$ (honest stake is finite, cannot double-count)

Adding the two settlement inequalities:

$$ES_H(r) + ES_H(r') + ES_B(r) + ES_B(r') \geq (4/3) * S_total$$

But:

$$ES_H(r) + ES_H(r') \leq S_H$$

$$ES_B(r) + ES_B(r') \leq 2 * ES_B_max = 2 * S_B / (1 + 0.9(k-1))$$

Therefore:

$$S_H + 2 * S_B / (1 + 0.9(k-1)) \geq (4/3) * S_total = (4/3)(S_H + S_B)$$

Rearranging:

$$S_H + 2*S_B/(1 + 0.9(k-1)) \geq (4/3)*S_H + (4/3)*S_B$$

$$2*S_B/(1 + 0.9(k-1)) - (4/3)*S_B \geq (4/3)*S_H - S_H$$

$$S_B * [2/(1 + 0.9(k-1)) - 4/3] \geq (1/3)*S_H$$

For $k \geq 2$ (any sybil attack with correlated identities):

$$2/(1 + 0.9(k-1)) \leq 2/(1 + 0.9) = 1.053$$

So:

$$S_B * [1.053 - 1.333] \geq (1/3)*S_H$$

$$S_B * (-0.280) \geq (1/3)*S_H$$

The left side is at most 0 and the right side is positive, so a coalition of $k \geq 2$ identities recorded with one shared profile, attesting alongside uncorrelated honest witnesses, cannot double-settle under the diversity-weighted predicate. This holds only for such a coalition: the adversary chooses how many identities it uses and what profiles they present, and with one identity (Step 5) or distinct profiles the discount does not apply. It also assumes that the diversity-weighted predicate gates finality, which the current implementation does not (Section 2.5).

Remark (shipped constants). Under the shipped runtime weights ($\gamma = 0.2$, so same-profile correlation reaches $\alpha + \beta + \gamma = 1.0$), the bound tightens: $ES_B \leq S_B / (1 + 1.0(k-1)) = S_B / k$, and the Step-4 coefficient becomes $2/k \leq 1 < 4/3$ for all $k \geq 2$, so the contradiction $S_B * [2/k - 4/3] \geq (1/3) * S_H$ holds a fortiori (the bracket is more negative than the spec-constant -0.280 for every $k \geq 2$). The discount is therefore larger under the shipped constants; the spec-constant derivation above is the smaller-discount case. The shipped γ additionally ramps with geographic diversity (0 below 2 distinct geo buckets, linear up to full γ at 12 witnesses — $\gamma_{effective_scaled}$); a smaller effective γ only lowers corr toward the spec value, staying within the two cases analyzed here.

Step 5: Uncorrelated adversary ($k=1$).

If the adversary uses a single identity ($k=1$), $d = 1$, $ES_B = S_B$. This reduces to classical BFT:

$$S_H + 2*S_B \geq (4/3)*(S_H + S_B)$$

$$S_H + 2*S_B \geq (4/3)*S_H + (4/3)*S_B$$

$$(2/3)*S_B \geq (1/3)*S_H$$

$$2*S_B \geq S_H$$

$$S_B \geq S_H/2$$

Since $S_total = S_H + S_B$, this means $S_B \geq S_total/3$. The classical BFT bound: the adversary needs at least $1/3$ of total stake.

Summary:

- **Safety bound:** below one third of the eligible stake, no adversary can double-settle, whatever its profiles (Theorem 1). This is the classical bound, as in PBFT.
- **At or above one third:** an adversary with one identity, or with identities recorded under distinct profiles, can double-settle; `spec/tla/MCTightBreak.tla` exhibits this.
- **Shared-profile coalition ($k \geq 2$):** loses effective stake to the discount; in the model above it cannot double-settle under the diversity-weighted predicate at any stake. Because the adversary chooses how many identities it uses and what profiles they present, this raises no bound against a rational adversary; it limits careless or honest concentration.

3.2.1 Slot Mutex: Operationalizing the Honest-Attestation Assumption

The proof of Theorem 1 relies on the premise that **honest identities do not attest conflicting records**. This is an assumption about operator behavior, but consensus cannot depend on behavior alone — it must be mechanized. The **slot mutex** (deployed in v0.7.6 of the reference implementation; see Protocol Whitepaper §11.12) is the mechanism that makes this assumption hold.

Lemma 3.1 (Slot Mutex Enforcement). Let r and r' be records with the same signed (`creator`, `nonce`) pair. Then no honest node will attest to both r and r' , even under adversarial delivery timing.

Sketch. Every v5+ record carries an ML-DSA-65 signature (Profile A records also a SPHINCS+ signature) covering a monotonic nonce drawn from the creator’s per-identity counter. Two records with the same (`creator`, `nonce`) therefore share a **slot key**

`slot_key(r) = hex(SHA3-256(creator_pk(r))) || ":" || hex16(nonce(r))`

(the shipped runtime materializes the key as a string — the lowercase-hex account hash, a `:` separator, and the nonce as 16 zero-padded lowercase hex digits — `record.rs::slot_key`). On first ingest of a record r , an honest node writes `slot_key(r) → r.id` in its slot index S . On subsequent ingest of any record r' with `slot_key(r') = slot_key(r)`, the node verifies a conflict proof $\pi = (r, r')$ — two valid signatures by the same creator over the same nonce — and, on success, writes `slot_key(r)` into its conflict index C . The attestation path consults C before signing: any record whose slot is in C is rejected from the attestation queue.

Because signature forgery is negligible under the standard ML-DSA-65 unforgeability assumption, the only way for a valid conflict proof to exist is for the creator itself to be Byzantine. An honest node cannot be coerced into attesting both sides of an equivocation by adversarial delivery ordering — it may temporarily attest whichever record arrives first, but once the second arrives, the slot enters C and settlement is blocked for both (see Protocol Whitepaper §11.12, “Settlement gate”). The settlement gate is local and pre-finality: a node that has marked a slot in C does not settle either record, whatever its attestation weight. It is not a hard floor. The mark is set only by a node that has seen both records; seal building and seal-member promotion do not consult C ; and a record finalized before the conflict is detected is not revoked. (Correction 2026-09-25: earlier editions called the gate a hard floor that closes the race where a Byzantine creator delivers r to a majority of validators before r' arrives. It does not close that race; against it, safety rests on the one-third bound of Theorem 1.)

Therefore the set of honest attestors to r and the set of honest attestors to r' are mutually exclusive, as Theorem 1 requires, and a node that has detected the conflict settles neither record. The slot mutex thus turns the “honest attestors are disjoint” premise into a structural invariant enforced by every honest node’s local state machine, not a behavioral assumption.

An earlier draft of the slot key included the record’s zone partition in the tuple, but this was rejected during conflict-injection testing: routing zones are derived from `record_id`, which differs per record by construction, so zone-keyed slots produced distinct keys for equivocating records and defeated the mutex. The canonical slot key is therefore (`creator`, `nonce`) only

— zone-independent. (Settlement itself is still evaluated in the zone derived from the record identifier, so two conflicting records can face different settlement zones; see the scope note after the proof of Theorem 1.)

3.3 Corollaries

Corollary 1.1 (Diversity Advantage). As predicates, diversity-weighted settlement implies classical settlement: any witness set that achieves diversity-weighted settlement also achieves classical settlement, but not vice versa. This does not reduce the adversary’s attack space in general: the current implementation gates finality with the classical predicate (Section 2.5), and an adversary that presents distinct profiles keeps its full stake under the discount.

Corollary 1.2 (Honest Diversity Requirement). For honest nodes to achieve settlement, they too must be sufficiently diverse. If all honest nodes share one profile (one org, one subnet):

$$ES_H = S_H / (1 + 0.9(h-1)) \quad \text{where } h = \text{number of honest witnesses}$$

(spec constants; S_H / h under the shipped corr = 1.0)

This creates an incentive for honest witnesses to diversify — exactly the behavior the protocol wants to encourage. Concentration of honest witnesses reduces the system’s ability to finalize records.

3.4 Concrete Examples

The following table shows the effective stake ratio for an attacker running k correlated nodes (all sharing the same org, subnet, and geographic zone — maximum correlation). Both constant conventions are shown: the paper’s spec constants (corr = 0.9, $d = 1/(1 + 0.9(k-1))$) and the shipped runtime constants (corr = 1.0, $d = 1/k$):

Attacker Nodes (k)	d — spec (corr 0.9)	Eff. stake — spec	d — shipped (corr 1.0)	Eff. stake — shipped
1	1.000	100%	1.000	100%
2	0.526	53%	0.500	50%
5	0.217	22%	0.200	20%
10	0.110	11%	0.100	10%
50	0.022	2.2%	0.020	2.0%
100	0.011	1.1%	0.010	1.0%
1,000	0.0011	0.11%	0.001	0.10%

(The $k = 5$ and $k = 10$ spec-constant entries correct arithmetic errors in earlier revisions of this table, which printed 0.238/24% and 0.122/12%; $1/(1 + 0.9 \cdot 4) = 0.217$ and $1/(1 + 0.9 \cdot 9) = 0.110$.)

The effective stake of a same-profile fleet decays as $1/k$ under the shipped constants. An attacker whose 100 nodes are recorded with one shared profile retains ~1% effective stake; the same attacker presenting 100 distinct profiles keeps its full stake, so the discount limits careless or honest concentration, not a deliberate adversary.

4. Post-Quantum Security

4.1 Threat Model

Let Sigma_d = ML-DSA-65 (FIPS 204) and Sigma_s = SPHINCS+-SHA2-192f (pre-standard; not FIPS 205 SLH-DSA) be the two signature schemes used in Profile A. Let A_q be a quantum polynomial-time adversary (Definition 22) that can break at most one of $\{\text{Sigma}_d, \text{Sigma}_s\}$ with non-negligible probability.

Theorem 2 (Profile A Record Unforgeability). Profile A records remain unforgeable under A_q . Because a forgery requires valid signatures under both schemes, the security level of a Profile A signature is about $\max(\lambda_d, \lambda_s)$, and in any case at least $\min(\lambda_d, \lambda_s)$.

Correction (2026-09-25) — scope of Theorem 2. Earlier editions titled this theorem “Dual-Algorithm Security”, stated that MESH-BFT *consensus* remains safe under A_q , and gave a composite level of $\lambda_d + \lambda_s$ (384 bits). Both claims were wrong. (1) Witness attestations carry a single signature field, verified with ML-DSA-65 only, and the settlement predicates read attesters’ stake and profiles, never their cryptographic profile. If ML-DSA-65 is broken, attestations — and therefore settlement — can be forged. Profile A protects record authorship, not consensus. (2) The work to break two schemes is the sum of the two works, not their product: for two 192-bit schemes, about $2^{192} + 2^{192} = 2^{193}$ operations, i.e. about 193 bits. The level is therefore about the stronger scheme’s, $\max(\lambda_d, \lambda_s)$, not $\lambda_d + \lambda_s$.

4.2 Profile A Unforgeability Proof

Step 1: Independence of hardness assumptions.

ML-DSA-65 relies on the hardness of Module-LWE (Learning With Errors over module lattices). SPHINCS+-SHA2-192f relies on the hardness of hash function preimage/second-preimage resistance (SHA-2 based).

These are independent mathematical problems:

- Module-LWE is a lattice problem. Best known quantum attack: BKZ with quantum sieving, achieving at most quadratic speedup over classical (Grover-like).
- Hash preimage resistance against quantum: Grover’s algorithm provides at most quadratic speedup ($2^n \rightarrow 2^{(n/2)}$).

Assumption: An oracle that breaks ML-DSA-65 provides no advantage in breaking SPHINCS+-SHA2-192f, and vice versa. This is a heuristic assumption based on the structural difference between lattice problems and hash preimage problems — no known reduction exists between them.

Step 2: Profile A signature security.

A Profile A record r has two signatures: $\text{sigma}_d(r)$ and $\text{sigma}_s(r)$. Verification requires BOTH to be valid:

$\text{Verify}_A(r) = \text{Verify}_d(\text{sigma}_d(r), \text{signable}(r), \text{pk}_d) \text{ AND } \text{Verify}_s(\text{sigma}_s(r), \text{signable}(r), \text{pk}_s)$

To forge a record, the adversary must produce valid $(\text{sigma}_d, \text{sigma}_s)$ for some chosen message m^* . Since A_q can break at most one scheme:

Case 1: A_q breaks Sigma_d . The adversary can forge sigma_d^* but cannot forge sigma_s . Therefore Verify_A fails because $\text{Verify}_s(\text{sigma}_s, m^*, \text{pk}_s) = \text{false}$.

Case 2: A_q breaks Sigma_s . The adversary can forge sigma_s^* but cannot forge sigma_d . Therefore Verify_A fails because $\text{Verify}_d(\text{sigma}_d, m^*, \text{pk}_d) = \text{false}$.

Case 3: A_q breaks neither. Standard unforgeability holds.

In all cases, Profile A records are unforgeable. QED.

Step 3: Profile B degradation.

Profile B uses only Sigma_d (ML-DSA-65). If A_q breaks Sigma_d:

- Profile B records become forgeable
- Profile A records remain secure (Sigma_s still holds)

This creates a trust boundary: Profile B identities are strictly less secure than Profile A under quantum attack.

Possible mitigation (not implemented): settlement could require a minimum fraction of attesting stake from Profile A identities — e.g., at least 50%. This would help only if attestations themselves carried both signatures, which they do not today (see the correction after Theorem 2).

Step 4: Security level.

For Profile A, a forger must produce valid signatures under both schemes. An attacker who can spend about 2^{λ_d} operations to break Sigma_d and about 2^{λ_s} to break Sigma_s can do both for about $2^{\lambda_d} + 2^{\lambda_s}$ operations, so the security level of a Profile A signature is

$$\lambda_{\text{MESH-A}} = \log_2(2^{\lambda_d} + 2^{\lambda_s}) \leq \max(\lambda_d, \lambda_s) + 1$$

For ML-DSA-65 (NIST Level 3, $\lambda_d \sim 192$) and SPHINCS+-SHA2-192f (NIST Level 3, $\lambda_s \sim 192$):

$$\lambda_{\text{MESH-A}} \sim 192 \text{ bits (at most 193)}$$

If one scheme is broken outright, a forger still needs the other signature, so the level stays at the unbroken scheme's level — about 192 bits here. This resilience to a single break, not extra bits, is what the second signature buys.

Correction (2026-09-25). Earlier editions multiplied the two success probabilities to obtain $\lambda_d + \lambda_s = 384$ bits as an “optimistic bound”. That multiplies probabilities at a fixed budget, but an attacker with a budget of 2^{192} operations breaks each scheme with probability close to 1; requiring both signatures adds about one bit, not 192.

4.3 Security Level Analysis

Configuration	Record signatures	Security of record signatures	If ML-DSA-65 is broken
Profile A	ML-DSA-65 + SPHINCS+-SHA2-192f	~ 192 bits	~ 192 bits (SPHINCS+ holds)
Profile B	ML-DSA-65 only	~ 192 bits	0 bits
Profile C	Delegated (inherits parent)	Depends on parent profile	Depends on parent profile
Witness attestations (all profiles)	ML-DSA-65 only	~ 192 bits	0 bits — settlement can be forged

For record authorship, Profile A survives the break of either scheme: a forger must break two different mathematical problems. The second signature buys this resilience, not extra bits; the

~192-bit level exceeds the 128-bit threshold conventionally considered sufficient for long-term security. Witness attestations do not have this property today (last row).

4.4 Profile B Trust Boundary

Under quantum attack on ML-DSA-65:

- All Profile B signatures become suspect
- Profile A record signatures remain secure (SPHINCS+ still holds)
- All witness attestations become forgeable, because every attestation carries only an ML-DSA-65 signature, whatever the attester's profile

Correction (2026-09-25). Earlier editions said the protocol enforces this trust boundary by letting settlement policy require a minimum Profile A quorum (for example, at least 50% of attesting stake), “since 50% of $2/3 > 1/3$ ”. No such policy exists in the implementation; attestations are single-signed in every profile, so a Profile A quorum would not help against a break of ML-DSA-65; and 50% of $2/3$ equals $1/3$, it does not exceed it. Dual-signed attestations, together with a Profile A quorum whose intersection bound is proven, are future work.

Note on mixed adversary bounds. The analysis above treats quantum adversaries (Sections 4.1–4.3) and Byzantine adversaries (Theorem 1) separately. A *mixed* adversary — one that controls Byzantine stake AND can break ML-DSA-65 — can forge attestations from any identity today, since attestations are single-signed (see the correction after Theorem 2). A bound for the mixed case requires dual-signed attestations and is left to future work.

4.5 Algorithm Migration

Corollary 2.1 (Algorithm Migration; design, not implemented). If one algorithm shows weakness (security level drops below threshold $\lambda_{\min}$), the network can:

1. Issue ALGORITHM_MIGRATION record with grace period
2. During grace period: both old and new algorithms accepted
3. After grace period: deprecated algorithm rejected
4. Records signed under the deprecated algorithm remain valid (cannot retroactively unsign)

Safety during migration: as long as $\geq 2/3$ of attesting stake has rotated to new keys, settlement remains safe. For Profile A record signatures, the dual-signature architecture removes the single point of cryptographic failure — one algorithm can be deprecated while the other continues to provide security. Witness attestations currently do have a single point of failure (ML-DSA-65).

5. Per-Record Causal Finality

5.1 Finality Depends Only on Witness Set

Theorem 3 (Per-Record Causal Finality). For any record r in a DAM M with at least 3 independent witness clusters whose combined stake $\geq (2/3) * S_{\text{total}}$:

1. r achieves Confirmed status in $O(1)$ attestation rounds
2. The confirmation latency is independent of $|V|$ (total records) and $|I|$ (total identities)
3. Causally concurrent records $r_1 || r_2$ finalize independently and in parallel

Proof.

Step 1: Finality depends only on witness set.

From Definition 16, Confirmed status requires:

$|W(r)| \geq 3$ AND $\text{clusters}(W(r)) \geq 3$ AND $\text{settled}_d(r)$

These conditions depend on:

- $|W(r)|$ — number of witnesses for THIS record
- $\text{clusters}(W(r))$ — cluster count for THIS record's witnesses
- $\text{settled}_d(r)$ — the diversity-weighted settlement predicate, computed from THIS record's witness set and the zone's registered stake denominator (a constant-size aggregate lookup)

None of these depend on $|V|$ (how many records exist in the DAM) or $|I|$ (how many identities exist in the network). They depend only on the local properties of r 's witness set, plus one $O(1)$ lookup of the zone's pre-maintained stake aggregate for the settlement denominator — which is a per-zone scalar, not a scan over records or identities. (In the current implementation, durable finality — the Finalized level — is reached when raw per-record settlement, Definition 18, passes, without the witness-count and cluster conditions; see the correction in Section 2.5. The same locality argument applies to that predicate.)

5.2 Attestation Round Complexity

Step 2: Attestation rounds are constant.

Define one “attestation round” as the time for a record to propagate to a witness and for the witness to submit an attestation back. This is bounded by:

$t_{\text{round}} = t_{\text{propagation}} + t_{\text{verify}} + t_{\text{sign}} + t_{\text{return}}$

where:

- $t_{\text{propagation}} = O(\log n / \log f) * t_{\text{hop}}$ with per-node fan-out f : $O(\log n)$ hops for constant fan-out, and a small constant number of hops (about 2) for $f \approx \sqrt{n}$ (the shipped fan-out is $\text{ceil}(\sqrt{p})$ for a node with p peers), at a cost of about $\sqrt{n}$ messages per node per hop (Correction 2026-09-25: earlier editions wrote $O(\log(n)/\log(\sqrt{n})) = O(\log n)$; that ratio equals 2)
- $t_{\text{verify}} = O(1)$ — ML-DSA-65 verification (~ 0.2 ms on a 2014 desktop CPU)
- $t_{\text{sign}} = O(1)$ — ML-DSA-65 signing (~ 0.7 ms on the same CPU)
- $t_{\text{return}} = O(\log n)$ network hops back

The NUMBER of attestation rounds to reach Confirmed is $O(1)$: witnesses holding $2/3$ of the eligible stake, across at least 3 clusters, attest in parallel once the record reaches them. Each round takes $O(\log n)$ hops of propagation for constant fan-out. Total confirmation latency:

$t_{\text{confirm}} = c * t_{\text{round}} = O(\log n)$ (c a small constant; $c = 1$ when all witnesses attest in parallel)

This is per-record latency. Leader-based BFT also commits in a constant number of rounds (HotStuff: three phases, with $O(n)$ messages per decision), so the advantage is not asymptotic latency: it is that independent records do not wait for each other or for a block.

Refined statement: Per-record finality requires $O(1)$ attestation rounds and $O(\log n)$ wall-clock time, where each attestation round involves gossip propagation. This is independent of $|V|$ (number of records) and depends only logarithmically on $|I|$ (number of nodes) through gossip propagation.

5.3 Parallel Finality for Concurrent Records

Step 3: Parallel finality for concurrent records.

Let $r_1 \parallel r_2$ (causally concurrent). Their witness sets $W(r_1)$ and $W(r_2)$ are computed independently:

- A witness l can be in both $W(r_1)$ and $W(r_2)$ — witnessing one does not prevent witnessing the other
- $T(r_1)$ is computed from $W(r_1)$ only; $T(r_2)$ from $W(r_2)$ only
- No ordering constraint between r_1 and r_2

Therefore r_1 and r_2 can finalize simultaneously, in parallel. The system's throughput for concurrent records is:

$\text{throughput} = \min(|\text{available_diverse_witnesses}|, \text{network_bandwidth}) * \text{records_per_round}$

In this analytical model, throughput scales linearly with the number of available independent witnesses, unlike block-based systems where throughput is bounded by block size and block time; it has not been measured at scale (Section 8.1).

Step 4: Comparison to block-based finality.

In a block-based system (PBFT, HotStuff, Tendermint):

- ALL transactions in block B must wait for B to be finalized
- Block finality requires communication among ALL validators
- Conflicting AND non-conflicting transactions are serialized into the same pipeline

In MESH-BFT:

- Each record finalizes independently
- Non-conflicting records never compete for consensus bandwidth
- The “block” equivalent is the individual record

Metric	Block-based (HotStuff)	MESH-BFT
Finality latency per tx	$O(\text{block_time} + c * \text{round_trip})$	$O(\log(n) * t_{\text{hop}})$, constant fan-out
Throughput	$O(\text{block_size} / \text{block_time})$	$O(\text{witnesses} * \text{records/round})$
Independent tx parallelism	None (serialized in blocks)	Full (finalize in parallel)

5.4 Throughput Scaling

Corollary 3.1 (Throughput Scaling). If the network has W diverse witness clusters, and each cluster can process p attestations per second, the theoretical throughput is:

$\text{max_throughput} = W * p$ (attestations per second)

Each record needs 3 attestations from 3 clusters. Therefore:

$\text{record_throughput} = W * p / 3$ (records confirmed per second)

This scales linearly with W (number of independent witness clusters), with no upper bound imposed by block size or block time.

5.5 Graceful Degradation

Corollary 3.2 (Graceful Degradation). If fewer than 3 diverse clusters are available:

- Records still achieve Attested status (1 witness, lower trust)
- No deadlock — the system continues accepting and propagating records
- When more clusters come online, records upgrade from Attested to Confirmed

Like block-based systems, MESH-BFT does not finalize a record until witnesses holding 2/3 of the eligible stake attest it: records stay Attested until then. In the current implementation, a network with fewer than three stakers also cannot seal epochs during a partition that excludes the genesis authority (Section 8.1). (Correction 2026-09-25: earlier editions called this strictly better than block-based systems.)

5.6 Bounded-View Liveness with Log-Depth Aggregator Chain

Sections 5.1–5.3 establish that a record’s confirmation latency is $O(\log n)$ *once an epoch seal is actually proposed by an honest aggregator*. That premise, however, is itself a liveness claim: an epoch must in fact be proposed and finalized within bounded time, even under Byzantine rank-0 proposers. This section formalizes the bound.

System assumptions (reiterated from Section 2).

- **Partial synchrony:** there exists an unknown global stabilization time GST after which message delays between any two correct identities are bounded by some Δ .
- **Honest-majority stake:** the Byzantine stake fraction within any zone is less than 1/3, and the global honest fraction is strictly greater than 2/3.
- **Fresh beacon:** the aggregator chain’s chained beacon $\text{beacon}(\text{zone}, \text{epoch}) = \text{SHA3-256}(\text{prev_seal_hash} \parallel \text{epoch} \parallel \text{zone})$ is produced from a seal hash the current proposer cannot pre-image under SHA3 collision resistance (this assumption does not hold in the current implementation: the previous sealer can grind the next beacon; see the correction after the proof).

Let $\text{base_timeout} = \text{clamp}(2 \cdot \text{p95_zone_rtt}, 1 \text{ s}, 10 \text{ min})$ be the adaptive per-zone timeout. Let $\text{MAX_VIEW_DEPTH} = 7$ bound the aggregator chain length.

Theorem 4 (Bounded-View Liveness). *Under the system assumptions above and at any time $t \geq \text{GST}$, every live zone produces and finalizes a seal for its current epoch within*

$$\begin{aligned} \text{finality_time}(\text{zone}, \text{epoch}) &\leq \sum_{k=0}^{\text{MAX_VIEW_DEPTH}-1} \text{base_timeout} \cdot 2^k + \Delta_{\text{cross}} \\ &= (2^{\text{MAX_VIEW_DEPTH}} - 1) \cdot \text{base_timeout} + \Delta_{\text{cross}} \\ &= 127 \cdot \text{base_timeout} + \Delta_{\text{cross}} \end{aligned}$$

where Δ_{cross} is the cross-zone round latency in the residual case where every rank in the per-zone chain is Byzantine. The probability that cross-zone escalation is needed is bounded above by

$$\Pr[\text{all MAX_VIEW_DEPTH ranks Byzantine}] \leq (1/3)^{\text{MAX_VIEW_DEPTH}} \approx 5 \cdot 10^{-4}$$

under uniform Byzantine sampling of the stake-weighted rank distribution.

Proof sketch.

Step 1 — beacon entropy. By the chained-beacon construction and SHA3-256 collision resistance, $\text{beacon}(\text{zone}, \text{epoch})$ is computationally indistinguishable from uniform over the output space to any proposer who has not yet observed the previous seal’s hash. Therefore the stake-weighted ordering produced by $\text{score}(\text{id}) = H(\text{beacon} \parallel \text{zone} \parallel \text{id}) / \text{isqrt}(\text{stake})$ sorts identities in a way no single proposer can bias at proposal time.

Step 2 — uniform Byzantine sampling. Let the zone’s staked set be partitioned into honest stake S_H and Byzantine stake S_B with $S_B / (S_H + S_B) < 1/3$. The priority score is monotone-increasing in a uniform hash, so for a *fixed identity partition* the probability that the rank- k position is occupied by a Byzantine identity is $(S_B^* / (S_H^* + S_B^*))$, the isqrt -damped effective-stake fraction. **Caveat (fragmentation): isqrt is concave, so it assigns proportionally more weight to small stakes; an adversary who splits S_B across many minimum-stake identities can therefore raise S_B^* relative to a monolithic honest staker — concavity does not by itself preserve the $< 1/3$ effective-stake bound.**

The bound holds only in conjunction with the minimum-stake admission floor ($\text{MIN_STAKE} = 100$ beats, enforced at record admission; Protocol Whitepaper §11.12), which caps the number of distinct Byzantine identities at $S_B / \text{MIN_STAKE}$ and thereby bounds the achievable isqrt -amplification. Under that floor the damped Byzantine fraction stays below $1/3$; a tight fragmentation-resistance bound as a function of MIN_STAKE is stated here as an assumption and left to the formal model.

Consequently, for each rank independently,

$$\Pr[\text{rank } k \text{ is Byzantine} \mid \text{previous ranks Byzantine}] < 1/3$$

and by chaining,

$$\Pr[\text{ranks } 0, 1, \dots, \text{MAX_VIEW_DEPTH} - 1 \text{ all Byzantine}] < (1/3)^{\text{MAX_VIEW_DEPTH}} = (1/3)^7 \approx 0.0004$$

This is a pessimistic bound — it treats rank occupancy as independent draws, while in practice the rank draws are negatively correlated under a fixed staker set (a Byzantine identity occupying rank k slightly decreases the probability another Byzantine identity occupies rank $k+1$). The true probability is therefore somewhat lower than $(1/3)^7$.

Step 3 — per-rank timeout soundness. By the exponential-backoff schedule

$$\text{unlock}(k) = (2^k - 1) \cdot \text{base_timeout},$$

at time $t = \text{epoch_start_ts} + \text{unlock}(k)$ rank k has had the interval $[\text{unlock}(k-1), \text{unlock}(k)] = [2^{(k-1)} \cdot \text{base_timeout} - \text{base_timeout}, 2^k \cdot \text{base_timeout} - \text{base_timeout}]$ of exclusive proposal authority (or 0 for $k = 0$). Under partial synchrony with message delay $\leq \Delta$ and $\text{base_timeout} \geq 2 \cdot \text{p95_zone_rtt} \geq 2 \cdot \Delta$ after GST, an honest rank- k proposer can (i) observe epoch_start_ts , (ii) construct a valid epoch seal, (iii) propagate it to $> 2/3$ zone stake, and (iv) collect attestations for finalization within rank k 's exclusive window — because the window doubles each rank while Δ remains fixed.

Step 4 — honest rank $\Rightarrow$ finalization within bounded time. Suppose rank k is the first honest aggregator, with all ranks $0, 1, \dots, k - 1$ Byzantine or crashed. By Step 3, once rank k 's window opens at $\text{epoch_start_ts} + \text{unlock}(k)$, an honest proposer seals and finalizes the epoch within rank k 's window of length $2^k \cdot \text{base_timeout}$. Total wall-clock bound for this case is

$$\text{finality_time} \leq \text{unlock}(k) + 2^k \cdot \text{base_timeout} = (2^{(k+1)} - 1) \cdot \text{base_timeout}.$$

Step 5 — summing over ranks. The worst per-zone case occurs when $k = \text{MAX_VIEW_DEPTH} - 1 = 6$ (only the last rank is honest):

$$\text{finality_time} \leq (2^{\text{MAX_VIEW_DEPTH}} - 1) \cdot \text{base_timeout} = 127 \cdot \text{base_timeout}.$$

Step 6 — residual case (all ranks Byzantine). In the measure- $(1/3)^7$ event that every rank is Byzantine, the zone is **stuck** at time $t = \text{epoch_start_ts} + 127 \cdot \text{base_timeout}$ and cross-zone escalation activates. A **global quorum seal** emitted by any staker in a different zone (with $\text{emitter_zone} \neq \text{stuck_zone}$) achieves finalization once $> 2/3$ of stake from non-stuck zones attests. By the system assumption that the global honest fraction strictly exceeds $2/3$ and the stuck zone contains at most all Byzantine identities, the non-stuck zones retain $> 2/3$ honest stake and escalation converges in one additional cross-zone round of duration

$$\Delta_{\text{cross}} \leq \text{p95_cross_zone_rtt} + 2 \cdot \Delta$$

which is a finite function of the mesh's inter-zone connectivity and adds a bounded offset to the finality bound.

Step 7 — combining. Taking the union bound over per-zone and cross-zone cases:

$$\begin{aligned} \text{finality_time} &\leq (2^{\text{MAX_VIEW_DEPTH}} - 1) \cdot \text{base_timeout} + \Delta_{\text{cross}} \\ &= 127 \cdot \text{base_timeout} + \Delta_{\text{cross}}, \end{aligned}$$

which establishes the claim. ■

Correction (2026-09-25) — Theorem 4 does not hold in the current implementation. Two defects break it. (1) *Seal settlement is not live.* With default configuration, a seal with two or more attesters cannot reach the diversity-weighted threshold, and a stuck seal does not escalate; records finalize through per-record settlement instead. More generally, seal settlement subtracts the proposer’s stake from the denominator and rejects the proposer’s own attestation, so an honest proposer holding a share p of the zone’s stake reaches $2/3$ against a withholding Byzantine share b only if $p \leq 1 - 3b$. With $b = 0.30$ and honest stakers of 0.30 , 0.20 and 0.20 , no proposer succeeds: a 0.20 proposer gets $3 \times 0.5 = 1.5 < 2 \times 0.8 = 1.6$, and the 0.30 proposer does worse. Step 3(iv) assumes those attestations arrive, and the escalation of Step 6 is skipped once any seal lands, finalized or not. In today’s single-staker network the eligible stake for seal attestation is zero, and no seal has settled. (2) *The beacon can be ground.* The previous sealer chooses fields that feed the next beacon and can search them to influence the next rank order. The $(1/3)^7$ bound assumes an unbiased draw and does not hold against such a sealer. In a single zone such a sealer can stall sealing; with several zones, escalation bounds the delay at $127 \cdot \text{base_timeout} + \Delta_{\text{cross}}$. The fixes are tracked: count the proposer’s stake as its vote, escalate on non-finalization, and derive the rank from a VRF. Until they ship, Theorem 4 and Corollaries 4.1–4.2 describe a design target with a proof sketch, not a property of the running system.

Corollary 4.1 (Typical-Case Finality). *In the honest-proposer case (rank 0 is honest), the finality bound collapses to $\text{finality_time} \leq \text{base_timeout} + O(p95_zone_rtt)$ — approximately one adaptive timeout plus one gossip round-trip. The shipped runtime sizes the timeout as $\text{base_timeout} = \max(2 \cdot p95_zone_rtt, 5 \text{ s})$ ($\text{src/network/zone_rtt.rs}$), so with the 5-second floor binding in a tight cluster ($p95_zone_rtt \approx 120 \text{ ms}$), honest-proposer finality is roughly 5–8 s wall-clock. A 1-second base_timeout (giving 1–3 s finality) is an illustrative lower-floor configuration, not the shipped value; lowering the floor is an operational tuning decision. Note the worst-case bound is symbolic — $127 \cdot \text{base_timeout}$ — and instantiates to $\approx 635 \text{ s} + \Delta_{\text{cross}}$ at the shipped floor, not “127 seconds”; it only materializes under sustained adversarial rank selection, which by Step 2 has probability strictly less than $1/3$ per rank and decays geometrically with depth.*

Corollary 4.2 (Independence from Zone Size). *The finality bound $127 \cdot \text{base_timeout} + \Delta_{\text{cross}}$ is independent of the number of identities in the zone. This is a direct consequence of bounding view depth at a constant $\text{MAX_VIEW_DEPTH} = 7$ rather than at $f + 1$: finality latency does not scale with stake concentration or staker count, preserving the $O(\log n)$ per-record latency of Theorem 3 even under Byzantine rank-0 proposers.*

Remark on depth selection. $\text{MAX_VIEW_DEPTH} = 7$ yields residual escalation probability $(1/3)^7 \approx 4.57 \cdot 10^{-4}$. Doubling depth to 14 would reduce this to $(1/3)^{14} \approx 2 \cdot 10^{-7}$ but would also raise the per-zone worst case from $127 \cdot \text{base_timeout}$ to $(2^{14} - 1) \cdot \text{base_timeout} \approx 16383 \cdot \text{base_timeout}$ — an unfavorable trade for a rare residual that cross-zone escalation already handles in $O(\Delta_{\text{cross}})$. Depth 7 is the knee of the curve: exponentially rare residual *and* bounded worst-case latency.

Remark on liveness slashing. Theorem 4 is a worst-case bound that assumes Byzantine rank-0 is *silent*. An actively hostile rank-0 that proposes malformed seals is trivially rejected at ingest (signature, Merkle-root, or zone-registry verification failure) and the rank advances immediately. Silent rank-0 is priced by the 1 %-stake liveness-failure slash (Protocol Whitepaper §11.12.3 Part D), which turns long-run silence from zero-cost to negatively-rational — no rational operator sustains rank-0 inactivity across multiple epochs. In the current implementation only

the genesis authority can emit this slash (Section 8.1), so the deterrent depends on that key.

6. Evaluation

Historical evaluation (March-April 2026). Every measurement in this section comes from the project’s first testnet, retired when its VPS fleet was decommissioned on 2026-06-09. Provider names and city locations are omitted per publication policy; node labels are generic (VPS-EU-1 ... VPS-US-2, Dev PC, Laptop). Quantities are stated in **beats**, the non-tradeable protocol credit (internal code symbol “TYOS” at measurement time; renamed the beat 2026-07), and the “faucet” was that era’s participation faucet (credits earned by running a node) — never a public token distribution. These are point-in-time historical measurements, not claims about a currently running network.

6.1 Testnet Deployment

We deployed a six-node geo-distributed testnet across 3 hosting environments (two commercial VPS providers plus residential machines) and 2 continents. The sixth node, a laptop behind NAT, was intermittent; where this section says “all 5 nodes”, it counts the five nodes online at the time of the check.

Node	Location	Hosting	Role	Latency to seed
VPS-EU-1 (seed)	EU (Central)	Commercial VPS	Genesis Authority + Anchor	—
VPS-EU-2	EU (North)	Commercial VPS	Witness	~20ms
VPS-US-1	US (West)	Commercial VPS	Witness	~150ms
VPS-US-2	US (East)	Commercial VPS	Witness	~120ms
Dev PC	EU (local)	Residential desktop	Anchor	~30ms
Laptop	EU (local)	Residential laptop	Witness (mesh VPN, NAT; intermittent)	~30ms

(Correction 2026-09-25: earlier editions listed five nodes and differed from the dataset in region labels and one latency figure; the table now follows the dataset.)

Network characteristics:

- EU-to-EU: ~20ms RTT
- EU-to-US: ~120-150ms RTT
- Gossip fan-out: $\text{ceil}(\sqrt{p})$ peers per hop for a node with p peers: 2 with five nodes online ($p = 4$), 3 with all six ($p = 5$)
- Maximum propagation: 2 hops to reach all nodes

At the time of measurement, the testnet had processed 15,381 records across 64 accounts with 2,401 beats staked. Supply conservation was verified at 10B beats (10^{19} base units) on all 5 nodes. Nodes maintained 14.9 hours continuous uptime with 3 active peers each.

6.2 Gossip Propagation Latency

Measured from the genesis node (VPS-EU-1) over 50 push measurements:

Percentile	Latency
p50	250ms
p90	500ms
avg	749ms

The average is skewed by 5 cross-Atlantic outliers exceeding 1 second. The EU-to-EU path (~20ms RTT) dominates the fast case; EU-to-US (~150ms) adds the tail latency. With $\sqrt{n}$ fan-out, propagation takes about 2 hops at any size ($\log n / \log \sqrt{n} = 2$), at the cost of $\sqrt{n}$ messages per node per hop.

End-to-end record finality timing (from a 60-second test session):

- Identity creation (PoW): ~10s desktop, ~30-60s phone
- Faucet claim: <1s
- First transfer completion: 3-4s
- Full flow (incognito browser to send): under 30 seconds

Cross-network operations verified:

- Transfer VPS-EU-1 to VPS-US-2: visible on all 5 nodes
- Transfer VPS-US-2 to Laptop: visible on all 5 nodes
- Transfer phone (Profile B) to laptop: received
- Proof VPS-EU-1 to VPS-US-2: verified (balance_range — the era log labeled this “Groth16”, incorrectly: the shipped construction was and is the Phase-1 SHA3-256 commitment scheme; Groth16 is design-stage and wire-rejected)
- Proof VPS-US-2 to VPS-EU-1: verified (content_commitment, same Phase-1 construction)

6.3 Projected Scaling

Using a gossip model with fan-out 3 per hop, we project confirmation latency as the network grows. These are projections, not measurements: the p50 column assumes 125 ms per hop, p90 twice that, and confirmation about four p50 propagation times. (Correction 2026-09-25: earlier editions labeled the hop column $\log N / \log \sqrt{N}$, which equals 2 for every N ; the column’s values are $\lceil \ln N / \ln 3 \rceil$, i.e. fan-out 3.)

Nodes (N)	Hops ($\lceil \ln N / \ln 3 \rceil$)	p50 Latency	p90 Latency	Confirmation (3 witnesses)
5	2	250ms	500ms	~1s
50	4	500ms	1s	~2s
500	6	750ms	1.5s	~3s
5,000	8	1s	2s	~4s
50,000	10	1.25s	2.5s	~5s

Under this model, confirmation latency grows as $O(\log n)$ with constant fan-out. At 50,000 nodes, the model estimates confirmation at ~5 seconds.

Under the analytical model of Corollary 3.1, throughput scales linearly with the number of independent witness clusters:

Witness Clusters (W)	Attestations/sec/cluster	Records Confirmed/sec
3 (2026-03 testnet)	100	100
10	100	333
50	100	1,667
100	100	3,333
1,000	100	33,333

The dataset records a measured rate of ~ 100 attestations/second per node. (Correction 2026-09-25: earlier editions attributed that rate to ~ 4 ms Dilithium3 verification per signature. The current ML-DSA-65 implementation verifies a signature in about 0.2 ms on a 2014 desktop CPU, so verification alone does not explain the rate; the limiting factor was not isolated.) The rows above 3 clusters are projections. The design has no block-size bottleneck.

6.4 Diversity Discount Impact

To illustrate the theoretical diversity discount, we compute the effective stake ratio for a correlated attacker with k nodes sharing the same organizational affiliation, network subnet, and geographic zone (maximum pairwise correlation; spec constants, $\text{corr} = 0.9$ — see Section 3.4 for the shipped-constant columns, $\text{corr} = 1.0$):

Attacker Nodes (k)	Independence $d(l_b)$	Effective Stake Ratio	Discount
1	1.000	100%	0%
2	0.526	53%	47%
5	0.217	22%	78%
10	0.110	11%	89%
50	0.022	2.2%	98%
100	0.011	1.1%	99%
1,000	0.0011	0.11%	99.9%

(The $k = 5$ and $k = 10$ rows correct arithmetic errors in earlier revisions, which printed 0.238/24% and 0.122/12%.)

The effective stake ratio decays as approximately $1/k$ for large k : a sybil fleet whose identities share one profile gains almost no effective stake from additional nodes. An attacker whose 100 nodes are recorded with one shared profile controls only 1.1% effective stake; the same attacker presenting 100 distinct profiles keeps its full stake (Section 8.1, Diversity oracle).

6.5 Zone Distribution

With $Z(r) = \text{SHA3-256}(\text{id}(r))[0]$ mapping records to 256 zones (the legacy first-byte assignment, in effect during the experiment; Definition 2 now specifies the dynamic full-entropy form) and 15,381 total records:

- Expected records per zone: $15,381 / 256 \sim 60$ (uniform distribution)
- SHA3-256 uniformity spreads honest records evenly. It does not stop a creator from choosing its zone: the creator chooses $\text{id}(r)$ and can re-draw it until the record lands in a target zone, at an expected cost of about N_z hash evaluations (Definition 2).

6.6 Conservation Invariant

The supply conservation invariant (beats) was verified continuously across all test scenarios:

- Normal operation: all 5 nodes report identical supply (10,000,000,000,000,000,000 base units = 10B beats)
- Node failure and restart: killed VPS-US-2, created records, restarted — VPS-US-2 caught up (14,644 records), supply consistent
- Genesis failure: killed VPS-EU-1 (genesis node) — network continued (VPS-EU-2 created records, VPS-US-1 received). Supply consistent on restart
- Cross-network transfers: no violation detected across 15,381 records processed

6.7 Comparison to Prior Systems

System	Throughput (TPS)	Finality Latency	Nodes Tested	Diversity Model	PQ-Safe	Per-Record
PBFT [1]	~1,000	1-5s	4-20	No	No	No
HotStuff [2]	~2,000	3-5s (3 rounds)	4-100	No	No	No
Narwhal/Tusk [5]	170,000	<3s	50	No	No	No
Bullshark [13]	130,000	<2s	50	No	No	No
Avalanche [3]	4,500	1-2s	2,000	No	No	Partial
Sui Lutris [7]	300,000	<500ms	100	No	No	Owned only
MESH-BFT (testnet)	~100*	~1s*	5 (of 6)	Partial†	Yes	Yes
MESH-BFT (projected)	~33,000*	~3s*	1,000	Partial†	Yes	Yes

*Estimated from measured attestation rates and projected gossip model. Actual benchmarks pending large-scale simulation.

†Computed and reported; finality is gated by raw stake in the current implementation (Section 2.5).

Our raw throughput is lower than Narwhal/Tusk and Sui Lutris because MESH-BFT does not batch records into blocks. The tradeoff is per-record granularity and the elimination of total ordering requirements. For applications that require per-record finality — sensor data, document signing, identity attestation, IoT provenance — this is the appropriate design point. Among the systems in the table, MESH-BFT is the only design that targets all three of diversity-weighted consensus, post-quantum signatures, and per-record finality; in the current implementation, diversity weighting does not gate finality (Section 2.5).

7. Related Work

7.1 Classical BFT

PBFT [1]. Castro and Liskov introduced practical BFT with a leader-based, three-phase protocol (Pre-Prepare, Prepare, Commit) achieving $O(n^2)$ communication in the normal case and $O(n^3)$ during view changes. Safety relies on quorum intersection — two sets of $2f+1$ replicas must overlap in at least $f+1$ honest nodes. PBFT is limited to approximately 20-100 nodes due to quadratic messaging, requires a designated leader, and has no model for correlated failures. MESH-BFT removes the leader and total ordering from per-record finality (epoch seals still use a ranked proposer chain, Section 5.6), and adds diversity discounting that has no PBFT equivalent.

HotStuff [2]. Yin et al. achieved linear BFT ($O(n)$ in both normal and view-change cases) via quorum certificates and a three-chain commit rule. The key innovation is that view changes are as cheap as normal operation. However, HotStuff remains per-block and leader-based, and threshold signatures require trusted setup. MESH-BFT's per-record finality is leaderless and has no threshold-signature dependency.

Tendermint [16]. Buchman formalized a BFT consensus engine for blockchain with immediate finality (no forks), partial synchrony, and $O(n^2)$ gossip-based communication. Tendermint powers the Cosmos ecosystem but shares the leader bottleneck and per-block finality limitations of classical BFT. MESH-BFT's gossip layer draws inspiration from Tendermint's approach but operates on individual records rather than blocks.

7.2 DAG-Based BFT

DAG-Rider [4]. Keidar et al. proposed a two-layer design where reliable broadcast builds a DAG and local interpretation produces total order with zero extra messages. Safety relies only on hashes, making it post-quantum safe as a side effect — though without a formal quantum adversary model. DAG-Rider achieves $O(n)$ amortized communication per transaction and is optimal in terms of message complexity. MESH-BFT differs in that we do not require total ordering and we formalize the post-quantum property of record authorship with an explicit QPT adversary model.

Narwhal and Tusk [5]. Danezis et al. separated reliable dissemination (Narwhal mempool) from ordering (Tusk consensus), achieving 170,000 tx/s with 50 nodes. Narwhal uses certificates of availability and worker-based scaling to achieve high throughput. MESH-BFT operates per-record rather than batching into blocks, sacrificing raw throughput for per-record granularity, and adds diversity and post-quantum models absent from Narwhal/Tusk.

Aleph BFT [6]. Gagol et al. designed a leaderless protocol with virtual voting derived from DAG structure and binary agreement via dual-threshold coin toss. Aleph is architecturally closest to MESH-BFT (leaderless, DAG-based, zero extra messages for ordering) but produces total order. It has no diversity model and no post-quantum analysis.

Bullshark [13]. Spiegelman et al. made DAG BFT practical by simplifying DAG-Rider's ordering rule while maintaining the same theoretical properties. Bullshark achieves 130,000 tx/s with sub-2-second finality on 50 nodes. Like all DAG-based protocols, it assumes independent Byzantine failures and requires total ordering.

7.3 Probabilistic Consensus

Avalanche [3]. Team Rocket introduced metastability-based consensus via repeated sub-sampled voting (sample size $k=20$). Avalanche is the closest system in spirit to MESH-BFT: it supports partial ordering, and independent transactions finalize in parallel without competing for consensus bandwidth. However, Avalanche provides only probabilistic safety (never absolute deterministic guarantees), uses sub-sampling rather than attestation accumulation, and has no diversity or post-quantum model. MESH-BFT provides deterministic safety under a one-third stake bound (Theorem 1), with diversity weighting as a confirmation signal.

7.4 Per-Object Finality

Sui Lutriss [7]. Blackshear et al. combined Byzantine Consistent Broadcast (for owned-object transactions requiring only $2f+1$ acknowledgements and sub-second finality) with full Mysticeti consensus (for shared-object transactions). Sui Lutriss achieves 300,000 tx/s for owned objects on 100 validators. The single-writer model is narrower than MESH-BFT's general DAG mesh, which supports multi-parent records. MESH-BFT additionally provides diversity weighting and post-quantum signatures.

FastPay [8]. Baudet et al. designed high-performance settlement for pre-funded payments using Byzantine Consistent Broadcast with $O(N)$ messages and single round-trip confirmation. FastPay is a predecessor to Sui with a much narrower scope (simple payment transfers only). MESH-BFT generalizes per-record finality beyond the payment use case.

7.5 Validator Diversity

GPoS [9]. The Geospatially-aware Proof of Stake protocol combines a Geographic Diversity Index (GDI) with stake to compute voting power, tested on HotStuff and CometBFT with a 45% improvement in Gini coefficient. GPoS is the closest prior art to MESH-BFT’s diversity model, but it provides no formal BFT safety proof incorporating diversity as a parameter — the GDI is evaluated empirically only. MESH-BFT defines diversity formally in its settlement predicate (Definition 19) and analyzes its effect on correlated coalitions (Section 3.2); the safety bound itself is the classical one-third bound.

Ethereum Validator Correlation Analysis [10]. Brown and Bautista-Gomez developed a correlation-adjusted index for measuring validator decentralization in Ethereum. This is a measurement and analysis contribution, not a consensus protocol modification. MESH-BFT operationalizes similar correlation insights directly in the consensus mechanism.

Flexible BFT [11]. Malkhi et al. introduced heterogeneous trust assumptions where clients select their own resilience model. “Diversity” in Flexible BFT refers to trust assumption diversity, not validator attribute diversity. MESH-BFT addresses a different dimension: the correlation structure of validator infrastructure.

Ethereum Client Diversity [14]. The Ethereum Foundation’s client diversity initiative addresses the engineering concern that validator software monoculture creates correlated failure risk. Ethereum incorporates diversity heuristically in slashing penalties but has no formal model. MESH-BFT formalizes this concern with Definitions 11-15 and analyzes its effect on correlated coalitions (Section 3.2).

7.6 Post-Quantum Blockchain

SoK: Blockchain Consensus in the Quantum Age [12]. Nash provides the most comprehensive survey of quantum-resistant consensus approaches (2019-2024). The survey catalogs existing defenses but does not propose a new adversary model — it identifies the gap that MESH-BFT addresses for record authorship with the QPT adversary (Definition 22) and the Profile A unforgeability proof (Theorem 2); consensus messages are not yet covered (Section 4.1).

QDBFT [17]. QDBFT uses Quantum Key Distribution (QKD) for BFT, requiring quantum hardware (QKD networks) rather than post-quantum classical cryptography. This is a fundamentally different approach — MESH-BFT uses classical post-quantum algorithms (ML-DSA-65, SPHINCS+) deployable on existing hardware.

8. Discussion and Future Work

8.1 Limitations

Current network: single staker. As of 2026-09-25 the public network runs one zone with one staked identity, the genesis authority. That identity is the only possible attester: its signature alone meets the 2/3 threshold for records it did not create, while its own records never settle, because a creator’s own stake is excluded from the denominator. Epoch seals are produced by the genesis authority under a bootstrap rule for networks with fewer than three stakers, and none has settled, because a proposer’s own stake is excluded as well. The network therefore

tolerates no Byzantine stake today, and the theorems of Sections 3–5 describe the design for many independent stakers, not the running network. Privileged actions — slashing and zone transitions among them — are accepted only from the genesis authority key; the code has no expiry for this power, and a slash does not require evidence of an offense (each slash is capped at 50%).

Implementation gaps against the model. Besides the corrections marked in Sections 2–5, four gaps are tracked as implementation fixes; all are latent or narrow in today’s single-zone, single-staker network: (1) attestations that arrive before their record are buffered and admitted later without the minimum-stake and identity-age checks; (2) with an active zone committee, the settlement numerator counts every attester while the denominator is committee stake; (3) the slot-conflict mark is local and pre-finality (Section 3.2.1); (4) the settlement zone is derived from the record identifier, so conflicting records can face different settlement zones.

Testnet scale. Our empirical results come from a six-node testnet (five nodes online during measurements). While the measured gossip latency (p50=250ms) and conservation invariant validation are informative, claims about behavior at 1,000+ nodes remain theoretical projections based on the $\sqrt{n}$ gossip model. Large-scale simulation is needed to validate these projections.

Simulation status. The safety-under-adversarial-conditions Monte Carlo *has* been executed: the tools/mesh-bft-sim harness sweeps a 64-combination adversary grid (network sizes {10, 50, 100, 500} × Byzantine fraction {0.10, 0.20, 0.30, 0.33} × sybil-cluster size {1, 5, 10, 50}) at the spec parameterization $\gamma = 0.1$ and records **zero safety violations** (committed results.json; re-validation at the shipped $\gamma = 0.2$ is pending). Liveness under churn and throughput under sustained load (Section 6.3) remain analytically modeled, not yet simulated. The sweep stops at a Byzantine fraction of 0.33, so it tests the regime Theorem 1 covers and cannot show anything above one third.

ZK curve limitation. The *specified* ZK proof subsystem targets BN254 (Groth16), which is not post-quantum secure — the bilinear pairing is vulnerable to quantum attacks via Shor’s algorithm on the discrete log problem. The shipped Phase-1 construction is SHA3-256 commitments (hash-based, no curve exposure; they do not yet hide a classified record’s content hash, see Protocol Whitepaper Section 5.3; the Groth16 wire version is rejected fail-closed until a prover/verifier lands), so this limitation becomes live when the zk-SNARK layer ships. Lattice-based ZK-SNARK constructions are an active research area; migration is planned when suitable schemes mature.

Profile B weakness. Profile B (single-signature, ML-DSA-65 only) is strictly weaker than Profile A under quantum attack. Browser and constrained-device nodes using Profile B create a heterogeneous security landscape. A Profile A quorum for settlement is not implemented, and witness attestations are single-signed with ML-DSA-65 in every profile (Section 4.1), so a break of ML-DSA-65 exposes settlement regardless of the profile mix.

Diversity oracle. The witness profile $P(I) = (\text{org}, \text{subnet}, \text{geo})$ relies on self-reported or inferred attributes. An adversary that can convincingly fake diverse profiles can circumvent the diversity discount. This is a bootstrapping problem: early networks with few validators have limited ability to verify profile claims. As the network grows, cross-validation and reputation mechanisms can harden the oracle.

Single-network scope. All definitions and proofs in this paper assume one network with one stake universe: the settlement threshold (Definition 18) is computed against stake registered in that network, and the safety argument (Theorem 1) bounds adversarial stake relative to it. Deployments that run multiple networks with disjoint roots of trust (e.g., permissioned or air-gapped instances) are sound only as **independent instances** of MESH-BFT — each with its own stake universe, settlement threshold, and security perimeter; the theorems make no claim across instances. Two consequences follow. (1) **No external time:** the protocol attests causal order; timestamps are operator-supplied and epochs use network-relative timeouts, so

nothing in this paper binds records to calendar time — externally-anchored state commitments are required wherever wall-clock claims matter. (2) **Cross-network history transfer is out of scope:** merging two long-disjoint DAGs (e.g., a private network publishing its history into a public one) has no conflict-resolution or finality-preservation semantics in this model. Until such semantics are formalized, the sound deployment rule is publication-as-attested-bundle: the receiving network attests the existence, identity continuity, and anchor trail of the imported bundle, but does not extend its own settlement guarantees over the bundle’s internal records, and imported history confers no stake, witness eligibility, or trust standing in the receiving network (see Protocol Whitepaper Section 10.6.3).

8.2 Future Work

Formal verification. Partial machine-checked verification is already in place and CI-gated: TLA+ specifications (`spec/tla/`) exhaustively model-check quorum-intersection safety and settlement liveness on bounded instances (the `formal-tla` CI job), and a ProVerif model (`spec/proverif/`, twenty-five scenarios) verifies the post-quantum handshake’s secrecy and authentication properties (the `formal-proverif` CI job). The TLA+ models use uniform stake and the full stake as the settlement denominator; they do not model creator or proposer exclusion, committee or zone denominators, correlated Byzantine sets above one third, or the raw-stake finality gate of the current implementation. MCTightBreak checks that the one-third bound is tight: with two of four equal-stake witnesses Byzantine, conflicting finalization occurs. What remains future work is extending the models to those gates and theorem-level machine-checked verification of the settlement bound (unbounded rather than bounded-instance) — a Lean 4 development is the intended next step.

Multi-network semantics. Formalizing cross-network history transfer: a merge model for long-disjoint DAGs with explicit conflict-resolution rules, finality-preservation statements across the publication boundary, and a safety theorem for the publication-as-attested-bundle rule described in Section 8.1. This is the consensus-side counterpart of the NETWORK_PUBLISH protocol (Protocol Whitepaper Section 10.6.3).

MPC ceremony for Groth16. The *specified* Groth16 subsystem requires a trusted setup (the shipped Phase-1 SHA3-256 commitments require none). When the prover/verifier lands, a multi-party computation ceremony with public verifiability would eliminate the trusted setup assumption.

Interval Tree Clocks. The current timestamp function `tau` uses `f64` timestamps, which are adequate for ordering but imprecise for causal tracking in distributed settings. Interval Tree Clocks [18] provide a more principled approach to causal ordering in dynamic distributed systems.

Lattice-based ZK-SNARKs. When post-quantum ZK proof systems (e.g., based on Module-LWE or hash-based commitments) reach practical maturity, adopting them in place of the specified BN254 Groth16 target will keep the proof layer fully post-quantum as it moves beyond the Phase-1 hash commitments.

Large-scale simulation. A Monte Carlo simulator covering 10,000-100,000 nodes under varying adversarial conditions (Byzantine fraction, sybil concentration, network partitions) is needed to validate the scaling projections and identify potential failure modes not captured by the analytical model.

Communication complexity. The $\sqrt{n}$ gossip model gives propagation bounds, but a formal analysis of total communication complexity for attestation accumulation — including bandwidth per node and message amplification — remains open.

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